Normal Distribution :-

One of the most important probability distributions is the Normal distribution. The continuous probability distribution with probability density function $f(x)$ given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where $-\infty < x < \infty$, $-\infty < \mu < \infty$ and $\sigma > 0$ is called the Normal distribution.

Note Mean and Standard deviation of Normal distribution are μ and σ respectively.

i.e Mean $= \mu$ and S.D $= \sigma$.

The Standard Normal distribution

The continuous random variable X which is distributed Normally has the probability function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

Here, $P(x_1 \le X \le x_2) = \int_{x_1}^{x_2} f(x)\,dx = \int_{x_1}^{x_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$

Putting $z = \frac{x-\mu}{\sigma} \Rightarrow x = \mu + \sigma z \Rightarrow dx = \sigma\, dz$

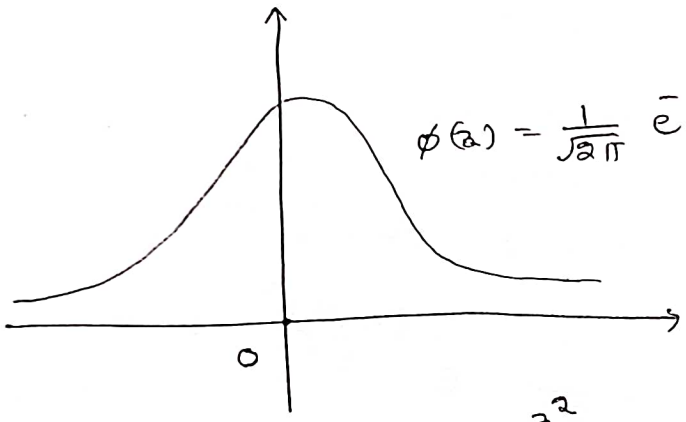
When $x = x_1$, let $z = z_1$ and when $x = x_2$, let $z = z_2$

i.e $z_1 = \frac{x_1 - \mu}{\sigma}$, $z_2 = \frac{x_2 - \mu}{\sigma}$

Then $P(x_1 \le X \le x_2) = \int_{z_1}^{z_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} \sigma\, dz$

$$= \int_{z_1}^{z_2} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$= \int_{z_1}^{z_2} \phi(z)\, dz, \text{ where } \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

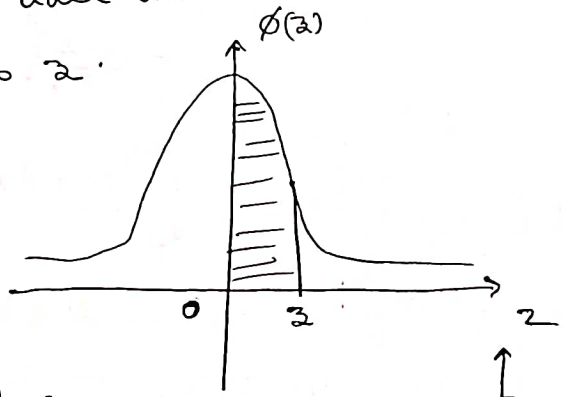


$\therefore$ The function $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$, $-\infty < z < \infty$ is known as the Standard Normal distribution, where $z = \frac{x-\mu}{\sigma}$.

we have $\phi(z) = \int_{z_1}^{z_2} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$. Further in particular if $z_1 = 0$, and $z_2 = z$

$$\phi(z) = \int_0^z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2} dz$$

This represents the area under the standard normal curve from $z = 0$ to z.

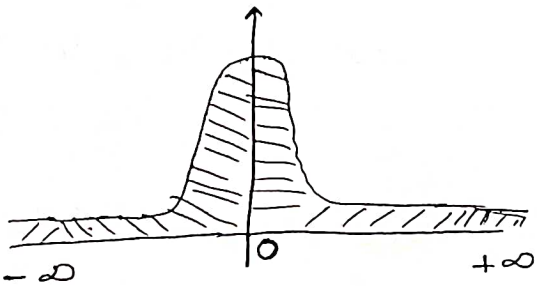


Area under Normal Curve:

(i) $\int_{-\infty}^{\infty} \phi(z)\, dz = 1$

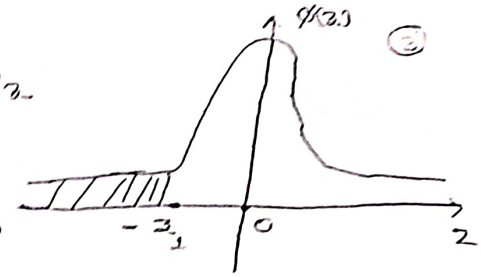
(ii) $\int_{-z_1}^{0} \phi(z)\, dz = \int_{0}^{z_1} \phi(z)\, dz$.

(iii) $\int_{-\infty}^{0} \phi(z)\, dz = \int_{0}^{\infty} \phi(z)\, dz = \frac{1}{2}$



(iv) $\int_{-\infty}^{-z_1} \phi(z)\,dz = \int_{-\infty}^{0} \phi(z)\,dz - \int_{-z_1}^{0} \phi(z)\,dz$

$= 0.5 - \int_{0}^{z_1} \phi(z)\,dz.$



Equivalently, we have-

(i) $P[-\infty \le z \le \infty] = 1$ (ii) $P[-\infty \le z \le 0] = 0.5$

(iii) $P[0 \le z \le \infty] = 0.5$ (or) $P[z \ge 0] = 0.5$

(iv) $P[-\infty < z < z_1] = P[-\infty < z \le 0] + P[0 \le z < z_1]$

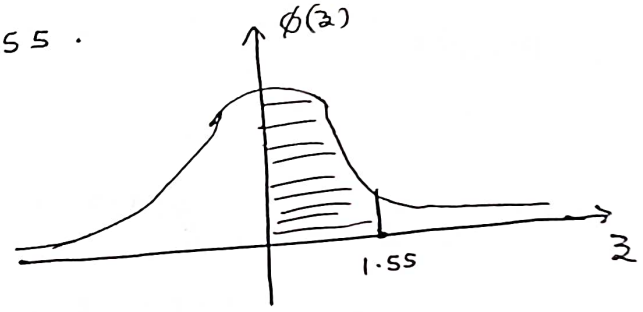
i.e $P[z < z_1] = 0.5 + \phi(z_1)$

(v) $P[z > z_2] = P[0 \le z \le \infty] - P[0 \le z < z_2]$

$= 0.5 - \phi(z_2)$

Illustration - 1

To find the area under the standard normal curve b/w $z = 0$ and 1.55.



$P[0 \le z \le 1.55]$

$= \phi[1.55] = 0.4394.$

Illustration - 2

Area of the standard normal curve between $z = -0.86$ and $z = 0$

$P[-0.86 \le z \le 0]$

$= P[0 \le z \le 0.86]$ (∵ By symmetry)

$= \phi(0.86) = 0.3501$

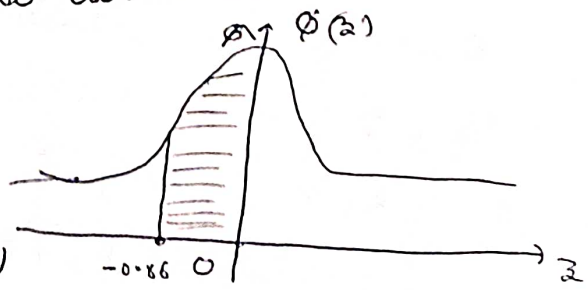
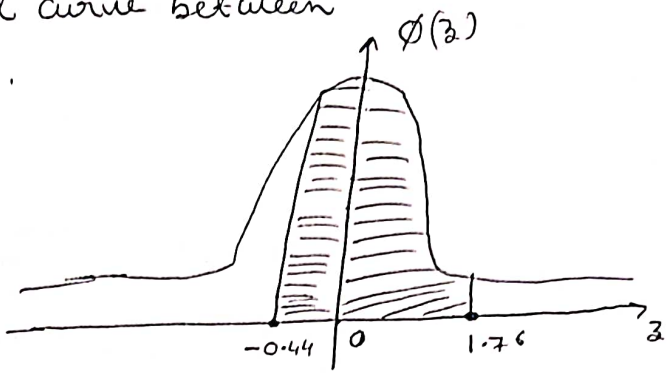


Illustration 3

Area of the standard normal curve between $z = -0.44$ and $z = 1.76$.



$P[-0.44 \le z \le 1.76]$

$= P[-0.44 \le z \le 0] + P[0 \le z \le 1.76]$

$= P[0 \le z \le 0.44] + P[0 \le z \le 1.76]$

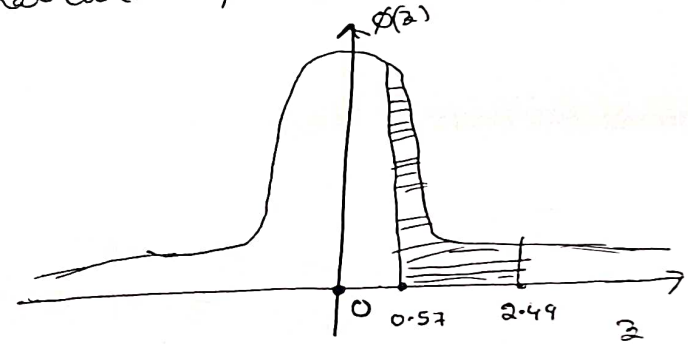
$= \phi(0.44) + \phi[1.76]$

$= 0.1700 + 0.4608 = 0.6308$

$\therefore P[-0.44 \le z \le 1.76] = 0.6308.$

Illustration 4

Area of the standard normal curve b/w $z = 0.57$ and $z = 2.49$.



$P[0.57 \le z \le 2.49]$

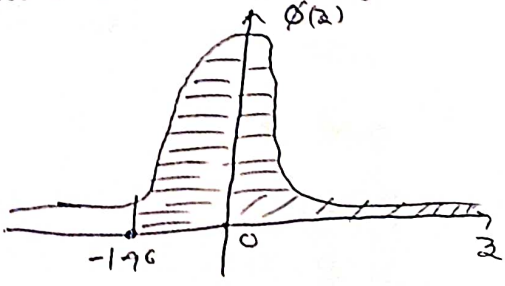
$= P[0 \le z \le 2.49] - P[0 \le z \le 0.57]$

$= \phi[2.49] - \phi[0.57]$

$= 0.4936 - 0.2157 = 0.2779.$

Illustration 5

Area of the standard normal curve to the right of $z = -1.96$



$P[-1.96 \le z \le \infty]$

$= P[-1.96 \le z \le 0] + P[0 \le z \le \infty]$

$= P[0 \le z \le 1.96] + 0.5$

$= \phi[1.96] + 0.5 = 0.4750 + 0.5 = 0.9750.$

Equivalently $P[z > -1.96] = 0.975.$

① · Evaluate the following probabilities with the help of normal probability table.

(i) $P[Z \geq 0.85]$ (ii) $P(-1.64 \leq Z \leq -0.88)$

(iii) $P(Z \leq -2.43)$ (iv) $P(|Z| \leq 1.94)$

▲ (i) $P[Z \geq 0.85] = P[0 \leq Z \leq \infty] - P[0 \leq Z \leq 0.85]$

$= 0.5 - \phi(0.85) = 0.5 - 0.3023 = 0.1977$

$\therefore P[Z \geq 0.85] = 0.1977$

(ii) $P[-1.64 \leq Z \leq -0.88] = P[0.88 \leq Z \leq 1.64]$ (∵ By Symmetry)

$= P[0 \leq Z \leq 1.64] - P[0 \leq Z \leq 0.88]$

$= \phi[1.64] - \phi[0.88]$

$= 0.4495 - 0.3106 = 0.1389$

$\therefore P[-1.64 \leq Z \leq -0.88] = 0.1389$

(iii) · $P[Z \leq -2.43] = P[-\infty \leq Z \leq 2.43]$

$= P[2.43 \leq Z \leq \infty]$ (∵ By Symmetry)

$= P[0 \leq Z \leq \infty] - P[0 \leq Z \leq 2.43]$

$= 0.5 - \phi[2.43]$

$= 0.5 - 0.4925 = 0.0075.$

(iv) · $P[|Z| \leq 1.94] = P[-1.94 \leq Z \leq 1.94]$

$= 2\,P[0 \leq Z \leq 1.94]$

$= 2\,\phi[1.94] = 2 \times 0.4738 = 0.9476$

② · If Z is a normally distributed with mean '0' and variance '1'; (i) find $P[Z \leq -1.64]$ and (ii) find Z_1, if $P[Z \geq Z_1] = 0.84$.

▲ Note: The distribution of Z is normal with mean Zero and S.d Equal to 1.

(i)

(i) $P[z \leq -1.64] = P[-\infty \leq z \leq -1.64]$

$= P[1.64 \leq z \leq \infty]$ (∵ By Symmetry)

$= 0.5 - \phi(1.64) = 0.5 - 0.4495 = 0.0505$

(ii) Given that $P[z \geq z_1] = 0.84$

Since $P[z \geq z_1] > 0.5$, if z_1 is -ve.

But, we have

$$P[z \geq z_1] = 0.84$$

$$P[0 \leq z \leq \infty] + P[0 \leq z \leq z_1] = 0.84$$

$$0.5 + P[0 \leq z \leq z_1] = 0.84$$

$$\phi[z_1] = 0.34$$

Now, refering the normal table for an area 0.34, we get $z_1 = 0.995$.

$\therefore z_1 = -0.995$.

③. If 'x' is a normal variate with mean 30 and standard deviation 5 find the probability that
(i) $26 \le x \le 40$ (ii). $x \ge 45$. (iii) $|x-30| > 5$

▲ We have standard normal variate

$$z = \frac{x-\mu}{\sigma}$$ Given that mean $\mu = 30$ and $\sigma = 5$.

$$z = \frac{x-30}{5}$$

(i) To find $P[26 \le x \le 40]$

At $x = 26$, $z = \frac{26-30}{5} = -0.8$

At $x = 40$, $z = \frac{40-30}{5} = 2$

Hence, we need to find $P[-0.8 \le z \le 2]$

$= P[-0.8 \le z \le 0] + P[0 \le z \le 2]$

$= P[0 \le z \le 0.8] + P[0 \le z \le 2]$

$= \phi[0.8] + \phi[2] = 0.2881 + 0.4772 = 0.7653$

$\therefore P[26 \le x \le 40] = 0.7653$.

(ii) To find $P[x \ge 45]$

At $x = 45$, $z = \frac{45-30}{5} = 3$

$P[x \ge 45] = P[z \ge 3] = P[3 \le z \le \infty]$

$= P[0 \le z \le \infty] - P[0 \le z \le 3]$

$= 0.5 - \phi(3)$

$= 0.5 - 0.4987 = 0.0013$.

$\therefore P[x \ge 45] = 0.0013$

(iii) $P[|x-30| > 5] = 1 - P[|x-30| \le 5]$

$= 1 - P[25 \le x \le 35]$

$= 1 - P[-1 \le z \le 1]$

$= 1 - 2P[0 \le z \le 1] = 1 - 2\phi(1)$

$= 1 - 2 \times 0.3413 = 0.6826$

At $x = 25$, $z = \frac{25-30}{5} = -1$

At $x = 35$, $z = 1$

④. If x is normally distributed with mean 12 and S.D 4, find the following:

(i) $P[x \geq 20]$ (ii) $P[x \leq 20]$

A We have S.N.V $z = \frac{x-\mu}{\sigma}$

Given that $\mu = 12$ and $\sigma = 4$

$$z = \frac{x-12}{4}$$

(i) We have $P[x \geq 20]$

At $x = 20 \Rightarrow z = \frac{20-12}{4} = 2$

$P[x \geq 20] = P[z \geq 2]$

$= P[2 \leq z \leq \infty] = P[0 \leq z \leq \infty] - P[0 \leq z \leq 2]$

$= 0.5 - \phi(2) = 0.5 - 0.4772$

$= 0.0228$

$\therefore P[x \geq 20] = 0.0228$

(ii) We have to find $P[x \leq 20]$

$P[x \leq 20] = P[z \leq 2]$

$= P[-\infty \leq z \leq 2] = P[-\infty \leq z \leq 0] + P[0 \leq z \leq 2]$

$= 0.5 + \phi(2)$

$= 0.5 + 0.4772 = 0.9772$

$\therefore P[x \leq 20] = 0.9772$

⑤. X is a normally variate with mean 30 and S.D 5, find the probability that

(i) $26 \leq x \leq 40$, (ii) $x \geq 45$ (iii) $|x-30| > 5$.

A We have S.N.V $z = \frac{x-\mu}{\sigma}$

Given that $\mu = 30$ and $\sigma = 5$

$$z = \frac{x-30}{5}$$

(i) To find $P[26 \le x \le 40]$

At $x = 26 \Rightarrow z = \frac{26-30}{5} = -\frac{4}{5} = -0.8$

At $x = 40 \Rightarrow z = \frac{40-30}{5} = 2$

$\therefore P[26 \le x \le 40] = P[-0.8 \le z \le 2]$

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(5). For a normally distributed variate with mean 1 and S.D 3, find the probabilities that

(i) $3.43 \le x \le 6.19$ [0.1644]

(ii) $-1.43 \le x \le 6.19$. [0.7686]

(6). The mean weight of 500 persons is 75 kg and the S.D is 7 kg. Find how many persons weight b/w 60 and 77 kg.

Given that $\mu = 75$ and $\sigma = 7$

$z = \frac{x-\mu}{\sigma} \Rightarrow z = \frac{x-75}{7}$

At $x = 60 \Rightarrow z = \frac{60-75}{7} = \frac{-15}{7} = -2.14$

At $x = 77 \Rightarrow z = \frac{77-75}{7} = \frac{2}{7} = 0.28$

$P[60 \le x \le 77] = P[-2.14 \le z \le 0.28]$

$= \phi[2.14] + \phi[0.28]$

$= 0.4838 + 0.1103 = 0.5941$

$\therefore$ No. of persons weight b/w 60 and 77 among 500 persons is $= 500 \times 0.5941 = 297$

(7). A Sample of 100 dry battery cells tested to find the length of life produced the following results $\bar{x} = 12$ hours and $\sigma = 3$ hours. Assuming the data to be normally distributed. what percentage of battery cells are expected to have life (i) more than 15 hours (ii) less than 6 hours (iii). b/w 10 and 14 hours.

A Let 'x' denote the length of life of dry battery cells. We have $z = \frac{x-\mu}{\sigma}$.

Given that $\mu = \bar{x} = 12$ and $\sigma = 3$.

$$z = \frac{x-12}{3}$$

(i) To find $P[x > 15]$

At $x = 15 \Rightarrow z = 1$

$$P[x > 15] = P[z > 1] = P[1 \leq z \leq \infty]$$
$$= P[0 \leq z \leq \infty] - P[0 < z < 1]$$
$$= 0.5 - \phi(1)$$
$$= 0.5 - 0.3413$$
$$= 0.1587$$
$$= 15.87\%.$$

(ii) To find $P[x < 6]$.

At $x = 6 \Rightarrow z = \frac{6-12}{3} = -2$

$$\therefore P[x < 6] = P[z < -2]$$
$$= P[-\infty \leq z < -2]$$
$$= P[2 \leq z \leq \infty]$$
$$= 0.5 - \phi(2) = 0.5 - 0.4772$$
$$= 0.0228 = 2.28\%.$$

(iii) To find $P[10 \le x \le 14]$

At $x=10 \Rightarrow z = \frac{10-12}{3} = -0.67$

At $x=14 \Rightarrow z = \frac{14-12}{3} = 0.67$

$P[10 \le x \le 14] = P[-0.6 \le z \le 0.67]$

$= 2P[0 \le z \le 0.6]$

$= 2 \times \phi[0.6] = 2 \times 0.2487$

$= 0.4974 = 49.74\%$

(8). In a test on 2000 electric bulbs, it was found that the life a particular make was normally distributed with an average life of 2040 hours and S.D of 60 hours. Estimate the no. of bulbs likely to burn for (i) more than 2150 hours (ii) less than 1950 hours and (iii) more than 1920 hours and but less than 2160 hours.

Given that $\mu = 2040$ and $\sigma = 60$

We have $z = \frac{x-\mu}{\sigma} = \frac{x-2040}{60}$

(i) To find $P[x > 2150]$

At $x = 2150 \Rightarrow z = \frac{2150-2040}{60} = 1.833$

$P[x > 2150] = P[z > 1.833] = P[0 \le z < \infty]$

$- P[0 < z < 1.833]$

$= 0.5 - 0.4664$

$= 0.0336.$

Thus the no. of bulbs expected to burn for more than 2150 hour among 2000 bulbs is

$= 0.0336 \times 2000 = 67$ approximately

(ii). To find $P[x < 1950]$.

At $x = 1950 \Rightarrow z = \frac{1950 - 2040}{60} = -1.5$

$P[x < 1950] = P[z < -1.5]$

$= P[-\infty < z < -1.5]$

$= P[1.5 < z < \infty]$

$= 0.5 - \phi(1.5)$

$= 0.5 - 0.4332 = 0.0668$.

∴ The no. of bulbs expected to burn for less than 1950 hours is $= 0.0668 \times 2000 = 134$.

(iii). To find $P[1920 < x < 2160]$

At $x = 1920 \Rightarrow z = -2$

$x = 2160 \Rightarrow z = 2$

$P[1920 < x < 2160] = P[-2 < z < 2]$

$= 2\,P[0 < z < 2]$

$= 2 \times \phi[2]$

$= 2 \times 0.4772 = 0.9544$

Thus the required no. of bulbs $= 0.9544 \times 2000$

$= 1909$

(9). In an Examination taken by 500 candidates, the average and S.D of marks obtained are 40% and 10% respectively. Assuming normal distribution, find (i) how many have scored above 60% (ii) how many will pass if 50% is fixed as the minimum marks for passing (iii) how many will pass if 40% is fixed as the minimum marks for passing and (iv) what should be the minimum percentage of marks for passing so that 350 candidates pass.

A Here $\mu = 40$ and $\sigma = 10$ in percentage.

we have $z = \frac{x-\mu}{\sigma} = \frac{x-40}{10}$

(i) For $x = 60$, we have $z = \frac{60-40}{10} = 2$

∴ The Probability that a candidate scores above 60% of marks is

$$P(x > 60) = P[z > 2] = P[2 < z < \infty]$$
$$= P[0 < z < \infty] - P[0 < z < 2]$$
$$= 0.5 - \phi(2) = 0.5 - 0.4772$$
$$= 0.0228.$$

Accordingly, the no. of candidates out 500 who have scored above 60% is

$$0.0228 \times 500 = 11.4 \approx 11.$$

(ii) For $x = 50$, we have $z = \frac{50-40}{10} = 1$.

∴ If 50% is fixed as the minimum marks for passing the probability that a candidate passes is

$$P[x \geq 50] = P[z \geq 1] = P[1 \leq z \leq \infty]$$
$$= P[0 \leq z \leq \infty] - P[0 \leq z \leq 1]$$
$$= 0.5 - \phi(1)$$
$$= 0.5 - 0.3413 = 0.1587$$

Accordingly, if 50% is fixed as the minimum marks for passing, the no. of candidates

out of 500 (who will pass) i.e

$$0.1587 \times 500 = 79.35 \approx 79.$$

(iii). For $x = 40$, we have $z = \frac{40-40}{10} = 0$.

∴ If 40% is fixed as the minimum mark for passing the probability that a candidate passes is

$$P[x \geq 40] = P[z \geq 0] = P[0 \leq z \leq \infty] = 0.5.$$

Accordingly, if 40% is fixed as the minimum mark for passing, the no. of candidates out of 500 who will pass is $0.5 \times 500 = 250$.

(iv) Let M% of marks be the minimum for passing if 350 of the 500 candidates are to pass. Then, we should have $M < 40$, and

$$500 \times P(x \geq M) = 350 \text{ (or) } P(x \geq M) = \frac{350}{500} = 0.7$$

$z = \frac{x-40}{6}$

$x = 10z + 40$

we note that

$$P(x \geq M) = P\{(10z + 40) \geq M\}$$

$$= P\left\{z \geq \left(\frac{M-40}{10}\right)\right\}$$

$$= P(z \geq -z_1), \text{ where } z_1 = \frac{40-M}{10} > 0$$

$$= P[-z_1 < z < 0] + P[0 < z < \infty]$$

$$= \phi(z_1) + 0.5$$

Since $P(x \geq M) = 0.7$

$$\therefore \phi(z_1) = 0.7 - 0.5 = 0.2$$

$\Rightarrow z_1 = 0.55$. Thus

$$0.55 = z_1 = \frac{40-M}{10} \Rightarrow M = 34.5 \approx 35.$$

Thus if 35% is fixed as the minimum mark for passing, then 350 candidates out 500 will pass.

(10) The marks of 1000 students in an examination follows a normal distribution with mean 70 and S.D 5. Find the no. of students whose marks will be (i) less than 65 (ii) more than 75 (iii) b/w 65 and 75.

Given that $\mu = 70$ and $\sigma = 5$.

We have $z = \frac{x-\mu}{\sigma} \Rightarrow z = \frac{x-70}{5}$

(i) For $x = 65$, $z = \frac{65-70}{5} = -1$.

$P[x < 65] = P[z < -1] = P[-\infty < z < -1]$

$= P[1 < z < \infty]$

$= 0.5 - \phi(1) = 0.1587$.

∴ no. of students scoring less than 65 marks

$= 1000 \times 0.1587 = 158.7 \approx 159$.

(ii). For $x = 75$, $z = \frac{75-70}{5} = 1$

$P[x > 75] = P[z > 1] = P[1 < z < \infty]$

$= 0.5 - \phi(1) = 0.1587$

∴ no. of students scoring more than 75 marks $= 1000 \times 0.1587 = 158.7 \approx 159$.

(iii) $P[65 < x < 75] = P[-1 < z < 1] = 2P[0 < z < 1]$

$= 2 \times \phi(1) = 2 \times 0.3413 = 0.6826$

∴ no. of students scoring marks b/w 65 and 75

$= 1000 \times 0.6826 = 682.6 \approx 683$.

(11). In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find the mean and S.D of the distribution.

▲ Let μ and σ be the mean and S.D of the normal distribution.

$P[x < 45] = 0.31$ and $P(x > 64) = 0.08$

We have $z = \frac{x-\mu}{\sigma}$

When $x = 45 \Rightarrow z = \frac{45-\mu}{\sigma} = z_1$ (say)

when $x = 64 \Rightarrow z = \frac{64-\mu}{\sigma} = z_2$ (say)

Thus, we have

$P(z < z_1) = 0.31$ and $P(z > z_2) = 0.08$

$\Rightarrow 0.5 + \phi(z_1) = 0.31$ and $0.5 - \phi(z_2) = 0.08$

$\phi(z_1) = -0.19$ and $\phi(z_2) = 0.42$

$\Rightarrow z_1 = -0.5$ and $z_2 = 1.4$

i.e $\frac{45-\mu}{\sigma} = -0.5$ and $\frac{64-\mu}{\sigma} = 1.4$

$\mu - 0.5\sigma = 45$ and $\mu + 1.4\sigma = 64$

By solving, we get $\mu = 50$, $\sigma = 10$.